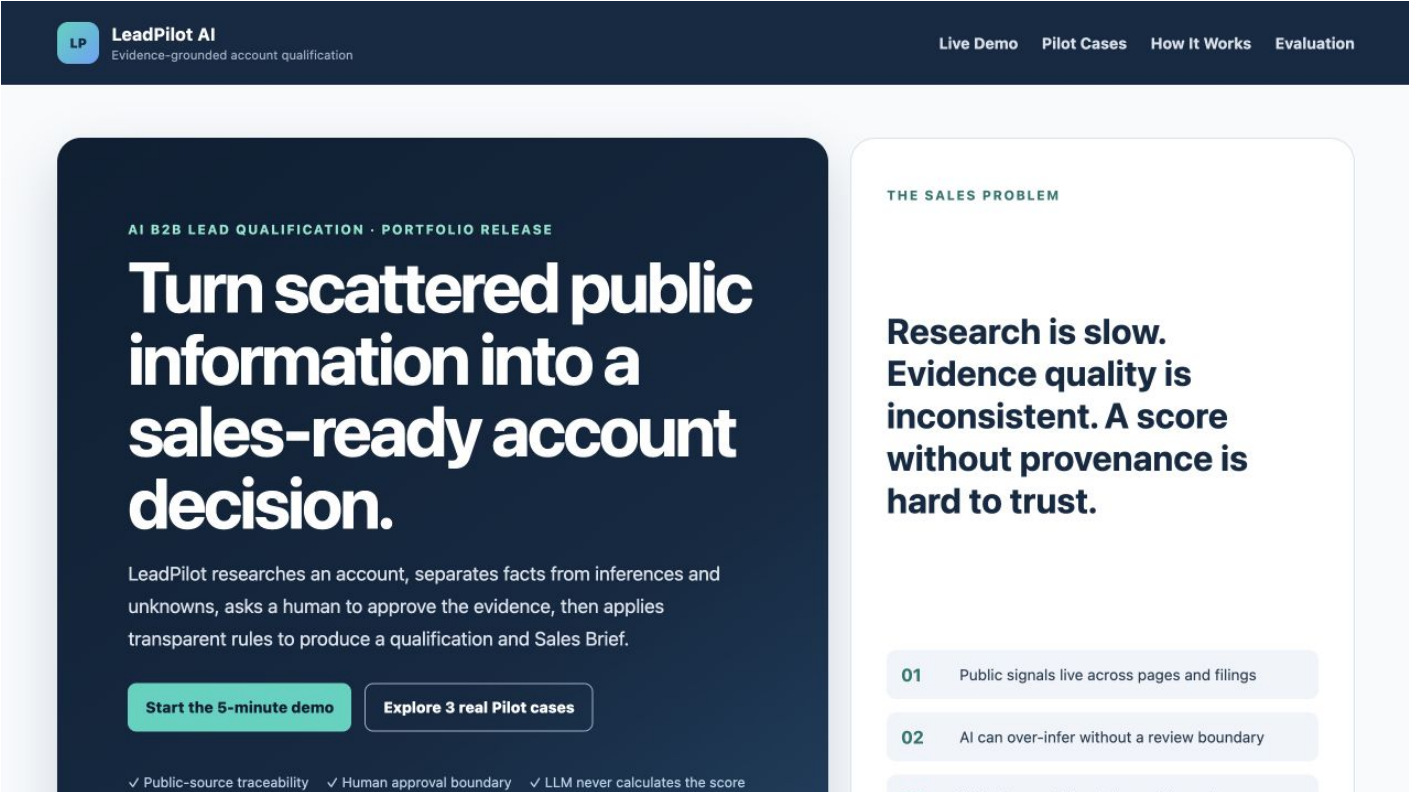


LeadPilot AI

Evidence-Grounded B2B Account Qualification & Sales Intelligence

AI 辅助 B2B 客户研究、资格判断与 Sales Brief 系统



Phase 7 portfolio home page · Business problem and five-minute demo entry

I designed a controlled workflow in which AI proposes evidence, a human decides what becomes formal Evidence, and frozen rules produce the account judgment.

3 REAL PUBLIC PILOTS	25 FORMAL EVALUATION	131 AUTOMATED TESTS	Frozen RELEASE STATE
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Business Problem

传统 B2B Account Research 的难点不是信息量，而是证据质量和判断可追溯性。

销售团队需要在官网、年报、新闻稿、支持页面和产品资料之间切换。研究结果常以零散笔记或摘要存在，事实、推断和缺失信息容易被混在一起。

Observed problem	Business consequence
公开信息分散	关键事实分布在多个来源，研究上下文难以复用。
人工研究慢	重复查找和整理占用销售前置判断时间，但本项目不据此宣称速度提升。
证据质量不一致	同一个结论可能来自直接事实、弱推断或没有找到的信息。
判断难以追溯	销售人员看到字母或摘要，却无法快速回到来源、审核动作和规则版本。

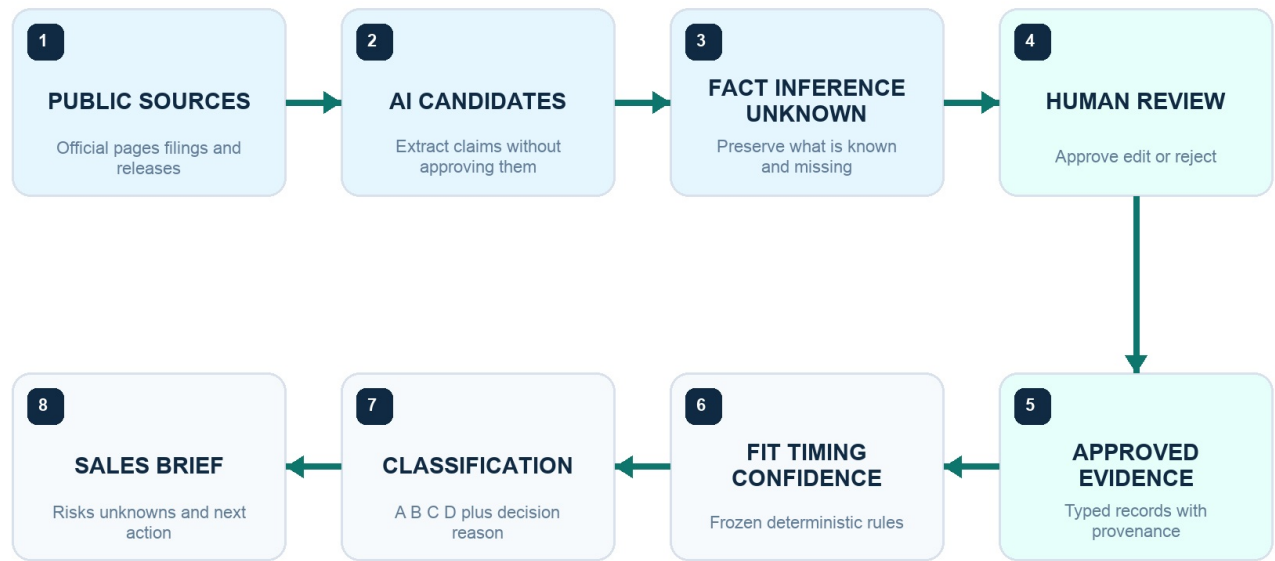
Product response

LeadPilot 将 Research、Evidence、Human Review、Scoring 和 Sales Brief 分离成可审计阶段。每个阶段保留输入、责任边界和失败条件。

- 公开来源先形成冻结 Research Snapshot。
- AI 只提出 Candidate，不批准 Evidence，也不计算评分。
- Fit 必须与 Evidence Coverage 同时出现，Unknown 不会被静默填充。
- 最终 Decision Reason 和 Sales Brief 可回到批准后的 Evidence。

Product Workflow

一条从公开来源到销售下一步的责任链。



Each stage owns one decision · Candidates cannot bypass Human Review

AI responsibility	Human responsibility	Rule engine responsibility
提出 source-grounded 候选并标注 F / I / U	批准、编辑或拒绝候选	读取 frozen rules.json 并计算结果

核心边界：LLM 不直接计算 Fit、Timing、Confidence 或 Classification。

ICP & Business Logic

三个目标账户类型与四个 VoltCare 服务场景。

ICP	Target account	Pilot	Evidence focus
ICP1	Smart EV brand / OEM	NIO Inc.	数字化售后、服务网络、远程诊断、数据接入
ICP2	Dealership / service network	Lithia & Driveway	网点规模、售后流程、配件和质保、决策归属
ICP3	Charging network operator	ChargePoint Holdings	充电运营、支持流程、API 能力、服务升级

VoltCare scenarios

Charging Failure	Parts Order	Warranty Support	Emergency Escalation
充电故障排查	配件订单查询	质保决策支持	高风险事件转人工

Business logic

- Fit 回答账户场景、规模、技术条件和组织可行性是否匹配。
- Timing 只使用有日期的事件信号；没有信号时明确显示 No timing signal found。
- Confidence 由 Coverage、关键 S1 Fact 和 unresolved conflict 共同决定。
- Classification A / B / C / D 表示账户研判，不等于 CRM Pipeline Status。

Human-in-the-loop

AI 提议 ; Human Review 决定哪些 Evidence 可以进入正式评分。

LP

LeadPilot AI
Evidence-grounded account qualification

Live Demo

Pilot Cases

How It Works

Evaluation

HUMAN APPROVAL BOUNDARY

REAL PILOT RECORDS

Human Review

AI proposed each candidate; a reviewer approved, edited or rejected it. Candidates cannot affect scoring before this boundary.

Showing: three real Pilot cases

Developer view: include Synthetic / MOCK records →

CANDIDATES
29

APPROVED
23

EDITED
3

REJECTED
3

REPRESENTATIVE DECISIONS

Review changes what the system is allowed to believe

These stored examples show the main human actions and the explicit treatment of uncertainty.

FACTAPPROVED

Source-grounded Fact

NIO reported 416 service centers worldwide as of December 31, 2025.

FACTEDITED

Human-corrected wording

NIO's privacy notice says app-displayed vehicle data is synchronized to a telematru service provider

INFERENCE
REJECTED

Over-inference stopped

NIO has approved funding to purchase VoltCare.

UNKNOWN
APPROVED

Uncertainty preserved

Do public materials establish an external API suitable for service and warranty-data integration?

REAL PUBLIC INFORMATION PILOT · Representative review outcomes from three stored Pilot runs

29
CANDIDATES

23
APPROVED

3
EDITED

3
REJECTED

- Approved Fact 保留 Source Level、URL、事件日期和原始 source span。
- Edited 记录 Agent 原文和 Human final claim , 不覆盖审核轨迹。
- Rejected over-inference 不会进入正式 Evidence 或影响评分。
- Unknown 被保留为待验证问题 , 并记录 searched sources。

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Portfolio Release

5

Account Evaluation

以 ChargePoint 真实公开信息 Pilot 展示 Evidence 到账户判断。

01 Sources

02 AI Candidate

03 Human Review

04 Score

05 Sales Brief

04

DETERMINISTIC BUSINESS JUDGMENT

Frozen rules calculate the score—not the LLM

CLASSIFICATION

B

Strong fit + limited current timing evidence

FIT

73.3

COVERAGE

80.0%

TIMING

30.0

CONFIDENCE

HIGH

Fit is always shown beside Coverage. Unknown is never treated as zero. The score reads only human-approved Evidence.

REAL PUBLIC INFORMATION PILOT · Frozen scoring v1.2 · ChargePoint Holdings, Inc.

73.3 FIT	80.0% COVERAGE	30.0 TIMING	HIGH CONFIDENCE
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Evidence	Rule output	Business interpretation
Human-approved Facts, Inferences and Unknowns	Classification B	Strong fit + limited current timing evidence

Decision Reason 与 Classification 分开保存。销售人员不仅看到 B，还能看到形成判断的业务原因和限制因素。

Sales Brief

把研究结果转成销售可使用的账户 Brief，同时保留 Unknown。

05 SALES BRIEF

Turn the decision into an actionable, traceable briefing

Why this account

ChargePoint publicly documents large network scale, APIs, guided app support, and support-case workflows.

The evidence supports a narrowly scoped integration discussion, while exact interface permissions and sponsorship remain

Key unknowns retained

Do available interfaces expose the exact charger telemetry and support-case fields required by VoltCare?

Who would own approval of an external service-

Next best action

Explore Pilot

No automated outreach or invented buyer contact.

ChargePoint Sales Brief excerpt · Every key conclusion links to approved Evidence

Brief section	What it gives the salesperson
Why This Account	说明场景匹配和证据依据，不复制一个孤立分数。
Key Evidence	链接回批准后的 Evidence、Source Level、日期和来源。
Key Unknowns	保留数据权限、商业 Sponsor 和决策归属等待验证问题。
Risks & Questions	明确不应追求的原因，以及 Discovery 需要验证的内容。
Next Best Action	给出 Explore Pilot、Research More 或 Validate Data Access 等建议。

系统不搜索联系人、不自动触达、不预测成交概率，也不把 Suggested Pilot 写成已测量 ROI。

Real-world Public Information Pilots

三个案例用于演示同一研究和审核流程；不构成企业评价或优先级排名。

Account	ICP	Fit	Coverage	Timing	Confidence	Class	Decision Reason
ChargePoint Holdings	ICP3	73.3	80%	30.0	HIGH	B	Strong fit + limited timing evidence
NIO Inc.	ICP1	66.7	55%	36.7	LOW	D	Insufficient evidence
Lithia & Driveway	ICP2	66.7	55%	26.3	LOW	D	Insufficient evidence

Pilot interpretation

- ChargePoint 具备较高 Coverage 和 High Confidence，但当前 Timing 信号有限，因此为 B。
- NIO 和 Lithia 的 Fit 并不低；关键维度 Coverage 只有 55%，Low Confidence 强制进入 D。
- D 是证据不足或冲突下的 Abstention，不是负面企业评价，也不是 ordinal zero。

Dataset	Count	Purpose	Display policy
REAL PUBLIC INFORMATION PILOT	3	Workflow demonstration	Default portfolio view
ENGINE VALIDATION DEMO	6	Synthetic engineering fixtures	Developer view only
FORMAL EVALUATION	25	Frozen Development / Test	Evaluation page only

Evaluation

冻结 Dataset、规则和 Official Test Run ，并保留负面结果。

LP

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Live Demo

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How It Works

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TRANSPARENT RESULTS · NOT ACCOUNT RANKING

FORMAL_EVALUATION

Evaluation & Limitations

Formal 25-Account Evaluation with frozen Development/Test governance, deterministic scoring and preserved negative results.

OFFICIAL TEST RUN · Cutoff 2026-09-30 · Dataset hash 9636d277165d3dd78a7e2ef3084b1f20389570a5187a86a7dd8a607a4ecbddc9
Recorded extractor evaluation; not a live-provider or independent-human generalization estimate.

WHAT THIS BUILD DEMONSTRATES

Workflow governance works end to end

- Frozen Test data and one official run are preserved.
- Agent Evidence and human Evidence remain separate.
- Scoring is reproducible, deterministic and append-only.
- Errors, D abstentions and missed A accounts remain visible.

WHAT IT DOES NOT DEMONSTRATE

No generalization or replacement claim

- Gold labels were not produced by an independent human panel.
- The extractor was recorded, not a live production LLM Provider.
- Timing measurements are not comparable.
- No ROI, productivity or deal-conversion conclusion is valid.

ACCOUNTS
25

DEV / TEST
8 / 17

H VS AGENT EXACT
64.7%

AGENT D ABSTENTION
5.9%

FORMAL EVALUATION · 25 accounts · Development 8 · Test 17 · One Official Test Run

H		HF	Agent
Pre-Agent Gold judgment		Frozen formula on Human Evidence	Frozen formula on Agent Evidence

Comparison	Exact	Cohen's κ	Interpretation
H vs HF	70.6%	0.4753	Rule logic does not fully reproduce Gold judgment
HF vs Agent	94.1%	0.8819	Extraction and reviewed Evidence are closely aligned
H vs Agent	64.7%	0.4035	End-to-end agreement remains limited

Negative result retained: A Precision 50.0% · A Recall 20.0% · Agent D Abstention 5.9%. The build validates governance mechanics, not generalization.

LeadPilot AI

Portfolio Release

9

Reliability & Guardrails

关键控制点集中在 Evidence 生命周期，而不是 Prompt 文案。

Control	Implementation	Risk addressed
Fact / Inference / Unknown	独立数据类型和 UI 标识	防止把推断或缺失信息伪装成事实
Evidence Provenance	URL、Source Level、日期、content hash	结论无法回到公开来源
Human Review	Approve / Edit / Reject + audit record	Agent candidate 直接污染正式 Evidence
Frozen Scoring Rules	Versioned rules.json	LLM 或结果导向调参改变数学规则
Confidence / Abstention	Conflict 和 Coverage 驱动 Low → D	低证据高 Fit 被当作强结论
Test Set Freeze	Gold first, then Agent; one Official Run	看到结果后反复调权重或标签
Audit Trail	Append-only ScoreResult and artifacts	历史结果被静默覆盖

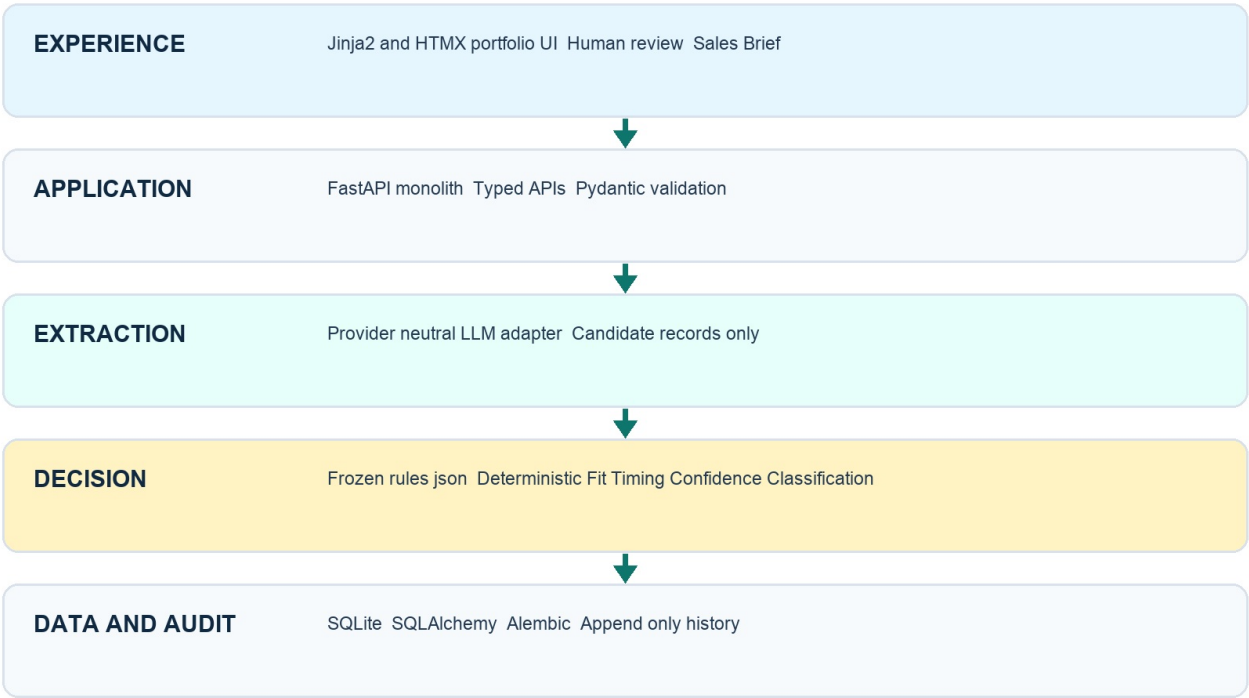
Evidence Sufficiency Check → Exclusion Gate → Fit → Timing → Confidence → Classification

Not found 不等于 does not exist。Exclusion 只有在满足证据要求的 Fact 存在时才能触发；证据不足继续评分并降低 Confidence。

Synthetic 和 Test 数据在 Dataset Purpose、API 与 UI 中严格隔离；默认 Portfolio 主路径只展示 Real Pilot。

Engineering Implementation

单体架构支持完整责任链，不以基础设施复杂度作为卖点。



Compact monolith · Provider-neutral extraction · Deterministic decision engine

Application	Data	Interface	Quality
FastAPI · Pydantic	SQLite · SQLAlchemy · Alembic	Jinja2 · HTMX	Pytest · 131 passed

LLM Candidate Extraction

Provider-neutral adapter 只写入 Candidate 层。Human Review 之后的 Evidence 才能被 frozen scoring v1.2 读取。LLM 不计算 Fit、Timing、Confidence 或 Classification。

Scope deliberately excluded: Kafka · Redis · Celery · vector database · automated outreach · contact search · ROI · complex CRM Pipeline.

Limitations

现有结果支持产品和治理演示，不能被扩张成商业效果或统计结论。

The current project cannot establish

- Agent 可以替代独立人工 B2B Account Prioritization。
- A 级账户在真实销售中具有更高转化率或商业价值。
- 系统已经产生真实 ROI、生产力提升或可比较的时间节省。
- 当前 25-account Evaluation 可以泛化到所有行业、地区或销售流程。

Why these claims remain open

Limitation	Observed boundary
Gold independence	Gold labels were evaluator-authored inside this build, not produced by an independent human panel.
Provider realism	The extractor is a recorded source-grounded run, not a live production LLM provider.
Timing comparability	Manual research, Agent run and Human Review measurements are not comparable.
Public evidence	Some integration, ownership and buying-intent fields are not publicly confirmable.

Negative Evaluation results and D abstentions remain visible. No Evidence, threshold, weight or Test label was changed to improve the portfolio story.

What I Designed

区分产品和业务设计责任，与 AI 工具辅助实现。

My product & business design	AI-assisted implementation support
问题定义和 Portfolio Scope	代码草拟、重构建议和重复实现任务
三个 ICP 与 VoltCare 业务场景	测试脚手架和边界用例补全
Evidence Schema 和 F / I / U 方法	UI 文案和版式迭代
Fit / Timing / Coverage / Confidence 逻辑	文档整理和格式转换
Human Review Workflow	调试辅助和实现选项比较
Evaluation Protocol 和 Test Freeze	在已定义规则内执行自动化检查
Failure Analysis 和 Limitations	生成可编辑交付物和演示素材
Product Boundary 和验收标准	不拥有业务规则、冻结决定或结果解释权

Authorship statement

我负责定义问题、业务对象、评分方法、审核边界、Evaluation Protocol、失败分析和最终 Scope，并对冻结规则与验收结果作出产品判断。AI 工具帮助实现和整理，但本作品集不暗示所有底层代码均由我逐行手写。

这一区分本身也是产品能力的一部分：能够使用 AI 加速实现，同时保留业务责任、验证标准和诚实披露。

Final Project Summary

适合简历和面试的一段项目总结。

7 PRODUCT PHASES	3 REAL PILOTS	25 FORMAL ACCOUNTS	131 TESTS PASSED
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LeadPilot AI 是一个 evidence-grounded B2B Account Qualification 与 Sales Research Agent。我设计了从公开来源、AI Candidate、Fact / Inference / Unknown、Human Review、Approved Evidence 到确定性评分和 Sales Brief 的完整产品链路。系统保留 Evidence Provenance、Unknown、Conflict、Confidence、Decision Reason 和 append-only audit history ；完成三个真实公开信息 Pilot ，以及一个冻结的 25-account Evaluation ，并原样保留 A Precision 50% 和 Recall 20% 等负面结果。

项目使用 FastAPI、SQLite、SQLAlchemy、Alembic、Jinja2 和 HTMX 实现可运行 Demo。LLM 只参与 Candidate Extraction ，不直接计算 Fit、Timing、Confidence 或 Classification。

Portfolio Freeze

Phase 7 是最终产品阶段。Phase 1–6 的代码、数据、Pilot、Official Evaluation、失败结果和 Limitations 均保留。后续只接受 Bug Fix ，不进入 Phase 8。

Designed for HR · Business Leaders · AI Product Managers · Customer Success & Solutions Roles